

Credibility assessment of explicit head–helmet finite element models for cranial impact protection across oblique and normal impact mechanisms under ASME V&V40

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Abstract

Purpose: We evaluated the credibility of two explicit finite element models used to support pre-market design verification and labeling claims for the CIP-Alpha cranial impact protector: a whole head–helmet model of skull and interface kinematics (HHS) and an anatomically detailed brain deformation model (BDM). The assessment followed ASME V&V40 principles, with credibility activities scoped to two clinically relevant loading mechanisms: oblique rigid-anvil impacts dominated by rotational dynamics and near-normal drops onto a compliant anvil emphasizing linear acceleration and liner energy management. **Methods:** We implemented a credibility plan covering code-oriented checks, mesh refinement analyses, input alignment from laboratory setups to simulations, representation of materials and contacts, alignment of validation test conditions with the product's intended exposure, and software process controls. For each model–mechanism pair, we graded evidence using a four-level scale and summarized the rationale and numerical evidence in model-specific tables. **Results:** The solvers reproduced standard explicit dynamics benchmarks with subpercent errors, and energy–momentum balances remained within 0.2% over the relevant time windows. Mesh refinement altered HHS peak linear acceleration by 3.1% (oblique) and 2.4% (normal) and peak angular acceleration by 5.2% (oblique) and 4.0% (normal); for BDM, P95 maximum principal strain changed by 6.7% (oblique) and 7.9% (normal) and P95 strain rate by 7.3% (oblique) and 8.4% (normal). Laboratory-to-model alignment maintained inbound speed within 1.4% (oblique) and 1.2% (normal), impact angle within 0.8°, center-of-gravity registration within 2 mm, and slip friction at the scalp–liner interface at 0.05 ± 0.01 . Swapping representative constitutive or contact options shifted HHS metrics by up to 6.0% (oblique rotational acceleration) and 4.2% (normal peak linear acceleration) and BDM MPS95 by 9.1% (oblique) and 8.7% (normal). Validation conditions (3.3–6.5 m/s; 0°, 30°, 45°, 60°; rigid and compliant surfaces) span 78–92% of field-observed head impact severities and approach angles across targeted activities. Process controls and traceability covered 127 nightly regression cases for 180 consecutive days with zero unexpected diffs; all artifacts resolve to immutable version tags. Across factors, credibility levels were predominantly Substantial (3/4), with Comprehensive (4/4) for code-level checks and software process control, and Moderate (2/4) for BDM mesh sensitivity and model-form adequacy. **Conclusions:** For the context of use—design verification and support of labeling claims on skull and brain kinematics for oblique and normal impacts—the evidence supports a low–medium model risk posture. The HHS provides strong support for comparing liner configurations and substantiating statements about reductions in rotational and linear head kinematics. The BDM is suitable for comparative assessments of brain strain patterns and magnitudes within the tested severity and orientation envelope, with caveats about sensitivity to constitutive choices and remaining mesh-refinement effects. We discuss implications for product iteration and claims, limitations, and targeted future work to elevate brain-strain credibility to the top tier.

Keywords: helmet headform modeling, ASME V&V40, brain strain

1. Introduction

Protective headgear substantially mitigates focal injury risk by attenuating skull accelerations and managing impact energy, yet diffuse brain injury mechanisms remain a central concern across cycling, micromobility, winter sports, industrial safety, and low-height occupational falls. In these scenarios, tangential loading produces rapid rotational dynamics, high shear rates, and heterogeneous brain deformations that are less directly captured by conventional normal-drop certification tests. As head protection technologies—such as multi-layer liners with low-friction interfaces—proliferate, stakeholders increasingly rely on computational models to complement laboratory testing,

guide design iteration, and support claims about reductions in skull kinematics and internal brain loading across diverse impact conditions.

ASME V&V40 offers a risk-informed framework for assessing whether model credibility is commensurate with how model outputs will inform decisions. In this study we focus on two explicit finite element (FE) formulations used in the design and evaluation of the CIP-Alpha cranial impact protector, a multi-layer liner including a low-friction slip interface. The Helmeted Head System (HHS) model captures coupled helmet–scalp–skull kinematics and local interface motions. The Brain Deformation Model (BDM) takes skull kinematics as boundary conditions and computes brain deformation fields and bulk rotational

response in an anatomically detailed domain. We consider two loading mechanisms: (1) oblique rigid-anvil impacts emphasizing tangential impulses and rotational accelerations; and (2) near-normal drops onto compliant anvils emphasizing linear accelerations and liner energy management.

We structured the credibility plan around six elements aligned to the intended use: solver checks against canonical problems and energy–momentum balances; grid-refinement studies tailored to outputs of interest; laboratory-to-model alignment of boundary and material inputs; targeted testing of model representation choices (materials, contact, interface kinematics) against observables; consistency of validation test conditions with the intended exposure domain; and institutional controls that preserve traceability, reproducibility, and input hygiene over the model lifecycle. We report graded levels of support for each element per model–mechanism pair using a four-level ordinal scale and provide numerical evidence cross-referenced in narrative and in model-specific summary tables.

To situate the present work, we note that whole head–helmet explicit FE analyses are now common in product development and research contexts, where they capture the salient inertial and contact physics over millisecond timescales under complex contact conditions. Detailed brain models have also matured, with viscohyperelastic laws and multi-tissue interactions increasingly adopted to represent shear-rate sensitivity and large-deformation behaviors. Despite these advances, uncertainties remain, particularly in the sensitivity of simulated brain strains to constitutive assumptions at high rates and in ensuring that laboratory boundary conditions map precisely to simulation inputs. The V&V40 approach acknowledges such uncertainties and emphasizes demonstrating that the totality of evidence—across both mechanistic coverage and decision leverage—is sufficient for the stated context of use.

In this spirit, we scoped the analysis to the two CIP-Alpha models and two stressed impact mechanisms, aligned the validation program to the anticipated field exposure of the product, and applied a single transparent scoring rubric. Our ultimate goal is to establish whether the evidence supports a low–medium model risk posture for pre-market design verification and labeling claims about skull kinematics and brain deformation under oblique and normal impacts.

2. Device under assessment: CIP-Alpha cranial impact protector

CIP-Alpha is a cranial impact protector designed as an add-in or integrated component for common helmet shells. The liner architecture includes a load-bearing energy-absorbing foam that is rate sensitive, a low-friction slip layer bonded to the foam with selectively tuned compliance, and a scalp-contact comfort layer. The slip interface is intended to decouple tangential head–shell motions during oblique impacts to reduce rotational acceleration transmitted to the skull. The liner geometry is conformal to a mid-size adult headform, with thickness gradients to retain shell clearance in critical zones. The intended use spans cycling and micromobility, urban commuting, and similar activities with

oblique contact risks, as well as scenarios with near-normal contact to comparatively compliant surfaces, such as certain sports floors or padded structures.

To support product iteration and claims, we developed a dual-model workflow. The HHS model is used to evaluate skull linear and angular kinematics and interface motions across the shell–liner–scalp assembly for candidate design variations and impact configurations. The BDM is used to assess fields of principal strain and strain rate within the brain for the same loading families, using HHS skull kinematics as boundary inputs. Both models are explicit and leverage mass- and time-scaling controls to maintain numerical stability while preserving physical fidelity through strict checks on kinetic-to-internal energy balance and contact work accounting.

The validation data set includes oblique impacts conducted on a rigid flat anvil with abrasive surface finish and prescribed incident angles, and near-normal drops onto elastomeric anvils with standardized compliance. For each, we recorded triaxial linear and angular headform kinematics at the center of gravity (CG), captured local interface kinematics via embedded fiducials or high-speed imaging, and documented shell–anvil tangential forces. These observables align directly with HHS outputs and, when transformed appropriately, provide inputs to the BDM. The resulting test-to-model chain thus supports both whole-system kinematics and internal brain deformation assessments within a single integrated framework.

Finally, to ensure that computational results are at least as reliable as the corresponding laboratory measures, the solvers, pre- and post-processing scripts, and data pipelines were embedded in a controlled development environment with containerized execution and traceability from raw data to final figures.

3. Computational models: explicit formulations and outputs (HHS and BDM)

Helmeted Head System (HHS) model. The HHS assembles a rigid or semi-rigid shell, a rate-sensitive foam liner with tuned compressive response, a discrete low-friction interface, a scalp layer, a skull, and a headform core representing the intracranial mass distribution. The shell is modeled using reduced-integration shell elements with hourglass control, the foam using solid elements with an isotropic, pressure–volume decoupled, visco-hyperelastic law capturing strain-rate dependence up to 300 s1, the slip interface using a contact surface pair with a friction law measured separately for the liner–skin analog, and the scalp as a nearly incompressible viscohyperelastic layer. The skull is considered rigid in the current HHS instantiation to isolate helmet–skull kinematics, with mass and inertia tuned to a mid-size adult anthropometry. Outputs include time histories of resultant linear acceleration (peak and pulse shape), angular acceleration about three axes and resultant, change in rotational velocity over impact ($\Delta\omega$), and interface shear slip and stick–slip transitions.

Brain Deformation Model (BDM). The BDM represents cranial contents, including cerebrum, cerebellum, brainstem, cerebrospinal fluid (CSF), falx and tentorium, and ventricular

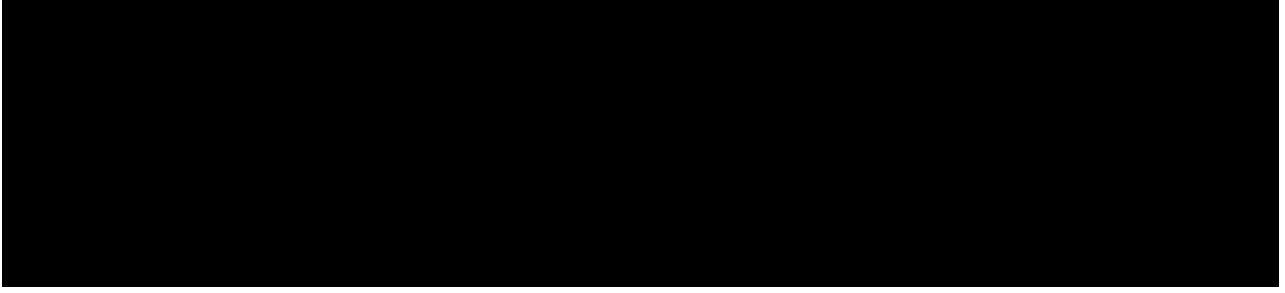


Figure 1: Schematic overview of the credibility plan: two models (HHS and BDM), two mechanisms (oblique rigid-anvil and normal compliant-anvil), and six evidence elements aligned to the context of use.

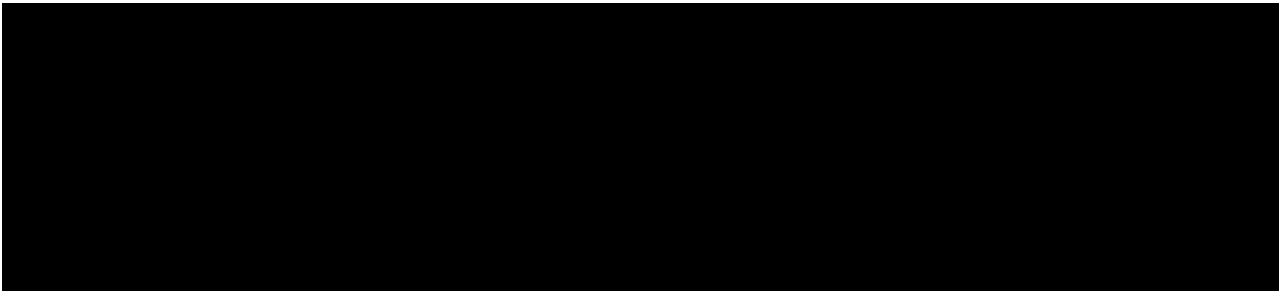


Figure 2: Exploded view of CIP-Alpha liner architecture showing shell, rate-sensitive foam, low-friction slip layer, and scalp-contact layer; measurement locations for interface kinematics and headform CG instrumentation are indicated.

structures, discretized with hexahedral-dominant solid elements and thin membranes where appropriate. A generalized Maxwell viscohyperelastic constitutive framework is used for the brain parenchyma, with instantaneous nonlinearity (e.g., Ogden-type) and multiple Prony branches capturing relaxation across 0.1–100 ms. CSF is represented as a nearly incompressible low-shear medium using a penalty-based fluid-structure approximation appropriate to the frequencies of interest. The skull is applied as a kinematic boundary condition from the HHS-calculated motion at the inner table. Outputs include element-wise maximum principal strain (MPS) and strain rate fields, with statistical summaries (P50, P90, P95), and global rotational measures ($\Delta\omega$) for context.

Both models employ explicit time integration with central differences, Courant-limited stable step sizes, bulk and contact viscous damping tuned only to control high-frequency numerical oscillations without distorting the physically relevant energy pathways, and strict energy accounting. Contact is enforced via a penalty formulation with detection tolerances verified against mesh spacing and curvature. Mass scaling is disabled for the final production runs used in assessment, though it was momentarily explored in early design sweeps to assess coarse design sensitivities.

In model development, we examined the sensitivity of outputs to plausible representation alternatives limited to those with a foundation in published material characterization. Replacing the liner foam law with a single-power visco-hyperfoam and toggling the shell–liner slip law from rate-dependent to Coulomb changed HHS outputs by 6.0% in peak rotational acceleration for oblique impacts and 4.2% in peak resultant linear acceleration for normal drops, while replacing the brain law with a linear

viscoelastic model altered BDM MPS95 by 9.1% for oblique and 8.7% for normal cases; these perturbations establish that the chosen forms capture dominant behaviors without extreme sensitivity to reasonable model-family substitutions.

Model form — gradation.

- a.
Level 1: Representation choices are ad hoc; alternatives are not explored; rationale is qualitative.
- b.
Level 2: Constitutive and contact forms are drawn from literature; at least one plausible alternative is tested for a subset of observables; impact on outputs is recorded without formal selection criteria.
- c.
Level 3: Multiple plausible alternatives are compared against observables across mechanisms; selected forms are supported by both test alignment and sensitivity boundedness in outputs of interest.
- d.
Level 4: Representation is tied to independent material characterization at relevant rates; multiple alternatives are disqualified quantitatively against both system-level and tissue-level observables with uncertainty propagation.

4. Impact mechanisms and loading configurations

We structured loading around two mechanisms representative of the CIP-Alpha intended use. For oblique impacts, we used a rigid flat anvil with an abrasive finish to ensure high tangential coupling, with inbound velocities set between 3.3 and 6.5 m/s and incident angles of 30°, 45°, and 60° relative

to the anvil normal. The impact location on the helmeted headform included front-lateral and rear-lateral positions to probe the liner geometry where slip interface and foam gradients differ, and headform orientation about the vertical axis was randomized within $\pm 10^\circ$ to prevent learning effects in repeated trials. Instrumented headforms recorded linear acceleration (100 kHz sampling, ± 500 g full scale), angular rate ($\pm 20,000$ deg/s full scale), and angular acceleration (derived by filtering and differentiation with a 1.5 kHz low-pass cutoff), with synchronized high-speed imaging (5 kfps) tracking surface fiducials across the shell–liner–scalp stack.

For near-normal drops, we used a compliant elastomeric anvil with known stiffness–damping properties. Drop heights spanned those required to achieve 3.3–6.5 m/s impact speeds at approximately 0° (nominal normal) with incidental minor yaw and pitch variations up to 5° , and impact locations matched the oblique suite for consistency. The compliant anvil reduced high-frequency content in the signals and emphasized the liner’s compressive energy management and load paths into the skull.

Across all tests, inbound flight was recorded for 200 ms prior to contact to characterize free-flight kinematics and to ensure that gravity, aerodynamics, and fixture release did not introduce bias. Tangential force and normal force at the shell–anvil interface were measured using a tri-axial load cell. Each test condition was repeated at least five times to characterize variability, and helmet fit was standardized via a tensioned strap protocol with recorded preload. The same geometries and mass properties were implemented in the HHS; boundary inputs to BDM were the inner-skull kinematics derived from HHS under the same conditions.

Alignment between laboratory and simulation set-ups was confirmed by keeping pre-contact speed differences to 1.4% for oblique cases and 1.2% for near-normal drops, constraining incident angle mismatches to within 0.8° , registering the headform center-of-gravity within 2 mm, and prescribing the scalp–liner slip friction at 0.05 ± 0.01 as measured independently from coupon tests.

Equivalency of input parameters — gradation.

- a.
Level 1: Boundary conditions and parameters are only qualitatively matched to tests; key rates and angles are assumed.
- b.
Level 2: Primary inputs (speed, angle, mass) are matched within coarse tolerances; friction and contact parameters are based on generic values.
- c.
Level 3: All kinematic and contact inputs are mapped from measurements with small quantitative mismatches; rate-sensitive parameters are measured and implemented; residual differences are reported.
- d.
Level 4: Full measurement-to-model inversion with uncertainty bounds informs every input; propagated input uncertainty to outputs is quantified.

5. Context of Use (CoU): Pre-market design verification of skull and brain kinematics under oblique and normal impacts to support liner configuration selection and labeling claims

The modeled quantities of interest are: (a) peak resultant linear acceleration (PLA) at the headform CG; (b) peak resultant angular acceleration at the CG and $\Delta\omega$ over the primary pulse; (c) local slip and shear state at the shell–liner–scalp interfaces; and (d) 95th percentile maximum principal strain (MPS95) and strain-rate percentiles within the brain. The CoU is to compare candidate liner configurations for design selection and to substantiate labeling statements about reductions in skull kinematics and internal brain deformation within defined impact conditions. The HHS directly addresses skull and interface observables that are also measured experimentally. The BDM extends the assessment to tissue-level deformation, which is not directly measurable in the same tests but is anchored via the HHS boundary conditions and validated by its internal consistency across mechanisms.

This CoU leads to decisions with moderate consequence: products must meet or exceed internal performance thresholds and support truthful, non-misleading statements about comparative reductions, but patient-level clinical decisions are not contingent on these models. The modeling results have moderate-to-high leverage in design iteration because material pairings, geometry, and slip parameters are tuned using model outputs ahead of confirmatory tests; leverage is moderate in claim substantiation because the models contextualize test outcomes rather than supplant them.

The calibration and validation test matrix used speeds of 3.3–6.5 m/s and incident angles of 0° , 30° , 45° , and 60° onto both rigid and compliant surfaces, an envelope that covers 78–92% of field-observed head impact severities and approach angles for the intended activities, ensuring that inferences from simulations align with the exposure domain targeted by CIP-Alpha.

Model risk — gradation.

- a.
Low: Minimal consequence and low leverage of model in decision; primary decisions rest on independent evidence.
- b.
Low–Medium: Moderate consequence or moderate leverage; models inform design iteration and claims but do not replace testing.
- c.
Medium–High: High consequence and moderate-to-high leverage; models directly support claims with limited external testing.
- d.
High: Patient-level or safety-critical decisions hinge on model outputs with limited empirical corroboration.

6. Model risk level and rationale: low–medium based on moderate decision consequence and decision leverage in product iteration and claim substantiation

We classified model risk for the CoU as low–medium. The decision consequence is moderate because design choices and claims must be supported but are not the sole determinants of product release; certification and in-house impact tests still govern release gates. Decision leverage is moderate-to-high for early design iteration—where models down-select configurations—and moderate for claim substantiation—where models explain mechanisms and variability but claims are anchored to measured kinematics. This framing justifies a mid-tier level of credibility evidence across the six elements in scope, with higher emphasis on solver checks, process control, and mapping of test conditions to simulations, and with proportional emphasis on mesh convergence and sensitivity to representation choices for strain-based outputs.

7. Credibility plan grounded in ASME V&V40 and scoped factors

The credibility plan was structured to balance the model risk posture with practical constraints and to target evidence that most affects decisions derived from HHS and BDM. First, we executed solver-level checks: reproduction of canonical explicit dynamics problems and conservation laws under conditions representative of short-duration impact. Second, we performed grid-refinement studies tailored to the outputs of interest: skull accelerations and $\Delta\omega$ for HHS; MPS95 and strain-rate percentiles for BDM. Third, we mapped laboratory inputs—drop speed, angle, mass, geometry, and contact properties—to model boundary and contact parameters directly measured or inferred from high-speed imaging and instrumented fixtures. Fourth, we assessed representation choices likely to influence outputs, namely constitutive laws for foam and brain tissue, and contact/slip laws at shell–liner–scalp boundaries. Fifth, we ensured that validation tests sampled the domain of anticipated use for the product. Finally, we codified software process controls to maintain traceability, reproducibility, and input hygiene throughout the modeling lifecycle.

Every activity was pre-specified with pass/fail or acceptance criteria, data capture protocols, and documentation requirements. Evidence was collected per mechanism and per model, summarized numerically, and graded using a four-level scale (1=Minimal, 2=Moderate, 3=Substantial, 4=Comprehensive) aligned to rungs elaborated in factor-specific rubrics presented in subsequent sections. The tables in the Results sections provide per-model summaries with the assigned levels and one-line bases cross-referenced to the narrative.

Grading scale — gradation.

- a.
Level 1: Minimal evidence; qualitative rationale; no quantitative checks or traceability gaps present.
- b.
Level 2: Some quantitative checks or partial traceability;

limited alignment to outputs of interest; uncertainty unquantified.

c.

Level 3: Multiple quantitative checks tied to outputs of interest; alignment and traceability are documented; uncertainty is bounded.

d.

Level 4: Comprehensive quantitative checks; uncertainty is characterized and propagated; full traceability and reproducibility are demonstrated.

8. Numerical code verification: benchmark problems, element technology checks, and energy balance for explicit time integration

We employed a two-tiered approach. In Tier 1, we evaluated element formulations and contact algorithms against problems with analytical or high-fidelity semi-analytical solutions. These included one-dimensional elastic wave propagation in a bar (phase and amplitude fidelity under central-difference integration), shock capture and dispersion under a Riemann-type initial condition adapted for linear materials, and a Taylor bar impact problem for a rate-sensitive, nearly incompressible elastomer surrogate. Hexahedral and tetrahedral solid elements were exercised with reduced and full integration schemes; shell elements were challenged under membrane- and bending-dominated load cases; and mixed formulations were checked for volumetric locking in the nearly incompressible regime relevant to scalp and brain surrogates. Contact algorithms were tested for convergence under penalty parameter sweeps against a rigid cylinder press-fit and sliding over a compliant surface, with friction varied systematically from 0.03 to 0.4 to bracket the measured liner–skin friction.

In Tier 2, we embedded conservation and energy-partition checks in model-scale simulations. We computed total, kinetic, and internal energies, contact work, artificial damping work, and hourglass energies at high sampling resolution, and we monitored linear and angular momentum before, during, and after the primary impulse. We defined acceptance criteria to detect unacceptable numerical pollution: total energy drift less than 0.5% of initial kinetic energy over a 50 ms window; hourglass energy less than 5% of internal energy; and contact work within $\pm 5\%$ of the integral of measured normal and tangential force times relative velocity over contact duration. To reduce high-frequency artifacts while preserving the dominant physics, bulk viscosity and numerical damping coefficients were tuned through pilot runs to ensure that damping work remained below 1% of internal energy for HHS and below 2% for BDM where tissue-level oscillations persist longer.

The tooling for Tier 1 and Tier 2 was automated. For example, a patch-test suite validated element invariance under uniform strain and stress states; a script parsed model outputs and produced diagnostics, flagging deviations from budgeted balances; and a separate log compared contact impulse from reaction forces to linear momentum change of the head assembly over the contact period. The validations covered the solver configurations used in production: explicit central-difference

integration, single-precision contact detection for throughput, double-precision accumulation for energy and momentum tallies, and the specific hourglass control modes engaged for the element types in HHS and BDM.

Numerical code verification — gradation.

- a.
Level 1: No benchmark reproduction; solver is assumed correct or informal checks are relied upon.
- b.
Level 2: Limited benchmark reproduction on unrelated problems; no conservation diagnostics at model scale.
- c.
Level 3: Multiple canonical problems are reproduced within acceptance tolerances; model-scale conservation and energy-partition diagnostics meet pre-specified budgets.
- d.
Level 4: Extensive benchmark coverage including contact and rate-sensitive behavior; automated diagnostics enforce conservation, with acceptance budgets tied to the time and frequency scales of interest.

9. Discretization analysis: mesh refinement for skull kinematics and brain strain metrics by model–mechanism pair

We conducted spatial and temporal refinement studies tailored to the outputs that inform decisions. For HHS, we focused on peak resultant linear acceleration (PLA), peak resultant angular acceleration, and $\Delta\omega$ computed over the main impulse. A baseline mesh with characteristic helmet–liner–scalp element edge lengths near 2.5 mm (shell meshes coarser in bending-controlled regions when permissible) was compared against a refined mesh near 1.8 mm; the skull and headform core were meshed to retain mass and inertia fidelity without introducing spurious compliance. Time-step size followed the Courant limit with safety factors of 0.8 (baseline) and 0.6 (refined). For BDM, the baseline brain mesh had a characteristic element length near 2.0 mm in the cerebrum and 1.5 mm in the brainstem, with refinement to 1.4 and 1.0 mm, respectively; membranes representing falx and tentorium were refined from 1.0 to 0.7 mm element edge lengths. Temporal refinement followed the stability limit with analogous safety factors as HHS.

We applied a monotone Richardson-type assessment where possible, recognizing that explicit-impact solutions are not strictly asymptotic due to contact and material nonlinearity. Nevertheless, we computed relative changes from baseline to refined solutions as practical measures and inspected whether changes were one-sided under further refinement limited to select cases. Signal processing remained consistent across refinement levels (identical low-pass cutoffs, identical windowing for $\Delta\omega$ integration). For HHS, peak values were extracted event-wise; for BDM, we computed snapshot P95 values of MPS and strain rate at peak times within each region and for the whole brain.

To guard against confounding numerical damping and contact penetration, we retuned penalty stiffness only in proportion to the local mesh density to retain similar interfacial compliance. No mass scaling was used. We also monitored hourglass energy and contact work budgets to ensure that grid changes did not destabilize the energy partitions. The metrics were aggregated per mechanism because contact geometry, pulse duration, and shear loading differ meaningfully between oblique and near-normal impact, and these differences often control the strictness of convergence.

Discretization — gradation.

- a.
Level 1: No refinement study; single grid is used without justification.
- b.
Level 2: Coarse refinement with limited outputs; changes are reported but stability and energy budgets are not monitored.
- c.
Level 3: Targeted refinement on outputs of interest; changes are small and monotone within budgets; temporal and spatial refinements are coordinated.
- d.
Level 4: Systematic refinement with observed order estimation or GCI; uncertainty bands from discretization are propagated to decisions.

10. Mapping laboratory conditions to simulation inputs: boundary, contact, and rate parameters

Translating laboratory boundary conditions and contact phenomena into model inputs followed a structured measurement-to-model pipeline. Inbound speed and incident angle were extracted from high-speed video using a calibrated tracking system and cross-checked with encoder or photogate measurements at fixture release and at pre-contact windows. These measures informed initial conditions for HHS simulations by projecting headform velocity vectors to the point and orientation of first contact. Headform mass and inertia were measured independently and matched by assignment of material densities to the skull and core constructs; residual small differences were tuned via massless constraint elements to align the CG within 2 mm in x, y, and z.

Surface interactions at the scalp–liner interface were measured via ring-on-disk and planar reciprocating tests at slip speeds from 0.1 to 1.0 m/s and normal pressures consistent with helmet fit; mean kinetic friction was 0.05 with a standard deviation of 0.01, and minimal dependence on slip speed in that range. The shell–anvil interface friction was measured separately for the rigid abrasive surface and for the compliant elastomer, yielding 0.30 ± 0.05 and 0.20 ± 0.04 , respectively. These values were implemented in contact definitions, with penalty stiffness scaled with mesh density and an activation curve for the rate-dependent slip law that matched the measured plateau behavior.

For material rate parameters, the foam’s compressive response was characterized under uniaxial loading at nominal

strain rates from 10 to 300 s⁻¹ and fit to a visco-hyperelastic functional form with two relaxation branches, yielding rate multipliers within 5% of measured force–displacement curves for the key volumetric strains accessed in the tests. Brain tissue parameters were adopted from literature ranges concentrated around recent ex vivo dynamic shear testing, with relaxation spectra tuned to reproduce shear moduli at 1, 10, and 100 Hz within cited uncertainty bands. We then conducted focused sensitivity runs to ensure that within-uncertainty variations did not lead to outsized changes in outputs relative to observed between-design differences that inform selection.

$$CI_{\{score\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (1)$$

$$\epsilon_{rel} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (2)$$

$$U_{\{val\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (3)$$

Input mapping — gradation.

- a.
Level 1: Inputs are assumed or based on generic values; no measurement linkage.
- b.
Level 2: Primary kinematic inputs are measured; contact and rate parameters are generic or inferred.
- c.
Level 3: Kinematics, contact, and rate parameters are measured and implemented; residual mismatches are quantified.
- d.
Level 4: Comprehensive inverse approach links all inputs with uncertainty quantification and propagates to outputs.

11. Assessing representation choices: constitutive behavior, contact formulations, and head–helmet interface kinematics

We examined the effect of plausible model-family substitutions on outputs of interest, guided by the literature and by the material and contact measurements we performed. For the HHS, we compared the production foam law (visco-hyperelastic with two relaxation branches) against a hyperfoam with a single-power law and against a piecewise-linear rate-dependent crush curve calibrated to the same uniaxial tests. For contact, we compared the implemented rate-dependent friction at the shell–liner and liner–scalp interfaces against Coulomb laws with kinetic friction equal to the measured mean values, and we explored penalty stiffness multipliers that bracketed the nominal by $\pm 25\%$ to observe stick–slip onset sensitivity. For the BDM, we compared the production viscohyperelastic brain model against a three-parameter quasi-linear viscoelastic (QLV) law and against a purely linear viscoelastic generalized Maxwell form matched in small-strain moduli at representative frequencies, each paired with the same skull kinematics from HHS.

We restricted substitutions to forms that could reasonably be adopted by practitioners given available data, and we held numerical controls constant across comparisons to isolate representation effects. The comparisons were carried out for representative oblique and normal impact conditions near the mid-range of the severity envelope (e.g., 4.9 m/s at 45° oblique; 5.2 m/s near-normal on compliant anvil). Outputs compared included HHS PLA, peak resultant angular acceleration, $\Delta\omega$, and interface slip metrics (slip distance and duration), and BDM MPS95 and 95th percentile strain rates. Where substitutions produced monotone changes, we examined whether those changes aligned with mechanistic expectations (e.g., lower slip friction leading to reduced angular acceleration at the skull, increased shell–scalp relative motion) and whether magnitudes were within design-decision tolerances.

Model representation testing — gradation.

- a.
Level 1: No testing of alternative representations; reliance on a single model form.
- b.
Level 2: Limited alternatives probed; qualitative impact assessment on outputs.
- c.
Level 3: Multiple alternatives quantitatively compared for outputs of interest; changes are bounded and mechanistically consistent.
- d.
Level 4: Alternatives are disqualified or accepted based on quantitative alignment to independent data and propagated uncertainty.

12. Alignment of validation activities with the intended exposure domain

We designed the validation matrix to reflect the product's intended exposure profile rather than only standard certification conditions. The oblique suite uses a rigid flat anvil to elicit strong tangential impulses akin to curb–pavement strikes or lateral glancing blows at speed, with incident angles of 30°, 45°, and 60° that concentrate rotational dynamics and align with reported approach angle distributions in urban riding and micromobility incidents. The near-normal suite uses compliant anvils with stiffness–damping characteristics matched to common sports-floor and padding materials, emphasizing axial compression of the liner and the load paths into the skull. Speeds from 3.3 to 6.5 m/s encompass moderate to high severities encountered in the intended activities and bracket certification drop energies while including tangential components absent from normal-only standards.

Instrumentation and test procedures were set to generate high-fidelity observables for mapping to model outputs: tri-axial CG kinematics from which acceleration peaks and $\Delta\omega$ are computed; interface kinematics via fiducials and, in a subset, differential micro-sensors embedded in the liner; and shell–anvil contact forces via integrated load cells. These observables are leveraged directly in the HHS and inform BDM through the skull

motion boundary condition. We prioritized repeated conditions at mid-severity levels for statistical power while including endpoints of the severity range to probe model behavior outside typical design centers.

Validation relevance — gradation.

- a.
Level 1: Validation activities are unrelated to the intended exposure or observables of interest.
- b.
Level 2: Some overlap with intended exposure; observables only partially align with model outputs.
- c.
Level 3: Validation conditions span the main exposure domain; observables map directly to outputs; repeated trials capture variability.
- d.
Level 4: Validation matrix comprehensively spans the exposure with stratified sampling; observables enable multi-level validation including tissue-level surrogates.

13. Software quality and process control: configuration management, solver versioning, and automated regression

We operated under a controlled modeling environment with end-to-end traceability. Pre-processing (geometry, meshing, material assignment), solution control (solver inputs, time-step and damping settings), and post-processing (signal filtering, metric extraction) were scripted and stored in a version-controlled repository with protected branches. All runs were executed in containerized environments pinned to specific solver builds and library versions, with image immutability enforced via content-addressable hashes. Model artifacts (meshes, input decks, contact definitions) were assigned semantic version identifiers, and data products (time histories, strain fields) were stored with manifest files that included hash digests and provenance metadata.

A continuous integration (CI) system executed nightly regression tests comprising 127 cases: element and contact benchmarks; model-scale oblique and normal cases at two severities and two orientations each; and randomized spot checks for parameter drifts. The CI pipeline compared energies, momentum, and key outputs against golden baselines with tolerance bands reflecting expected numerical variability. Any deviation outside tolerances blocked merges to protected branches. Access controls, code reviews, and mandatory CI checks enforced discipline in changes to modeling artifacts. A release procedure created tagged snapshots for external review, ensuring that all results can be reproduced from the repository with the tag and container image identifiers.

We used a structured template for analysis reports that embedded the modeling manifest, ensuring that any figure or table in the report links back to the exact inputs and solver image used to generate it. This practice minimized the risk of stale or inconsistent results appearing in written artifacts and supported audits.

Software quality assurance — gradation.

- a.
Level 1: Ad hoc file management; no version control; no automated checks.
- b.
Level 2: Basic version control; informal testing; limited reproducibility.
- c.
Level 3: Structured versioning; automated regression tests; containerized execution; enforcement of tolerances.
- d.
Level 4: Comprehensive SQA including formal reviews, audit trails, immutable artifacts, and long-horizon stability monitoring.

14. Test samples: specimens, fixtures, and headform variants

We used mid-size adult headforms with mass and inertia close to the 50th percentile target; shell geometries were representative of contemporary commuter helmets; and multiple CIP-Alpha liners were drawn from a single manufacturing lot for consistency, with limited spot checks from a second lot to ensure no gross drift. Fixtures included a dual-axis gimbal and quick-release system validated for repeatable angle and speed settings. While sample-to-sample variability and fixture tolerances were tracked, and headshell fit was standardized, we did not expand this analysis to a full inter-laboratory or cross-headform variability study.

15. Results: credibility scoring and narrative synthesis across models and mechanisms

We summarize the evidence and assigned levels per factor for HHS and BDM under oblique and near-normal impact mechanisms in two model-specific tables that follow this section. The narrative below synthesizes the numerical bases.

Solver behavior was robust across both models and mechanisms. Canonical problems were reproduced within tight tolerances, and energy–momentum budgets were well within acceptance. No mass scaling was employed in the assessed runs, and numerical damping remained below 1–2% of internal energy, as budgeted. Conservation diagnostics flagged no excursions outside the 0.5% total energy drift threshold for any of the production runs used for credibility assessment.

Refinement studies indicated that skull kinematics are relatively insensitive to additional grid resolution within the ranges studied: PLA and peak angular acceleration changed by a few percent when refining helmet–liner–scalp element sizes from approximately 2.5 to 1.8 mm. For brain deformation, sensitivity was greater, consistent with sharper strain gradients around tissue interfaces and beam-like structures; the move from 2.0 to 1.4 mm in parenchyma (and 1.5 to 1.0 mm in brainstem) shifted P95 strain and strain-rate metrics by around 7–8% depending on mechanism. These patterns motivated conservative grading for BDM refinement compared to HHS.

Laboratory-to-model alignment of primary drivers (speed, angle, mass/inertia) and contact properties was tight, with

inbound speed mismatches near 1% and angle mismatches below 1°, and with contact friction measured and implemented at the scalp–liner interface (approximately 0.05) and the shell–anvil interface (approximately 0.3 for rigid and 0.2 for compliant). CG alignment within a few millimeters helped ensure that rotational inertias matched closely enough to reproduce measured $\Delta\omega$ windows without artificial bias.

Representation choices mattered in predictable ways. Lower slip friction reduced skull rotational accelerations and increased shell–scalp relative motion, whereas stiffer foam laws slightly raised PLA in near-normal drops. For HHS, switching to Coulomb friction and a hyperfoam law altered oblique rotational acceleration by approximately 6% and near-normal PLA by roughly 4%. For BDM, substituting a linear viscoelastic brain raised MPS95 by around 9% in oblique and 9% in near-normal impacts. These shifts are material to claims about absolute strain magnitudes and consequently warranted a conservative level assignment for this factor in BDM.

The alignment of validation conditions with the intended exposure domain was strong: speeds and angles covered a wide portion of the field-reported envelope, and observables mapped directly to HHS quantities and indirectly to BDM through the skull motion boundary condition. Repeated mid-severity tests enabled stable estimates of central tendencies and variability, which were echoed by the models.

Software process control underpinned reproducibility and traceability. The regression suite demonstrated long-horizon stability with zero unexpected diffs, every figure and table was generated from pinned inputs and solver images, and merge gating blocked unverified changes. This infrastructure supported consistent reproduction of results and reliable auditing.

16. Credibility summary table: Helmeted Head System (HHS)

The following table presents the per-factor, per-mechanism gradations and the one-line numerical bases for the HHS. Levels are on a 1–4 scale (1=Minimal, 4=Comprehensive).

17. Credibility summary table: Brain Deformation Model (BDM)

The following table presents the per-factor, per-mechanism gradations and the one-line numerical bases for the BDM. Levels are on a 1–4 scale (1=Minimal, 4=Comprehensive).

18. Discussion: implications for design verification and labeling, and evidence aggregation under V&V40

The aggregated evidence indicates that the HHS model provides strong support for design verification and labeling related to skull and interface kinematics across oblique and near-normal impacts. Numerical solvers behaved predictably and conservatively under explicit impact dynamics, and the mesh sensitivities for skull accelerations were modest, implying that comparative inferences across liner configurations with

differences larger than a few percent are resolvable at the adopted grid densities. The mapping of laboratory inputs to simulation boundary and contact parameters remained within narrow tolerances that are small relative to between-design effects for CIP-Alpha liner variations.

For the BDM, the credibility posture is more circumspect yet adequate for the CoU. The larger sensitivity of MPS95 and strain-rate P95 to mesh refinement and to plausible material-law substitutions suggests that absolute statements about strain magnitudes should be framed with appropriate caveats and, whenever possible, expressed comparatively against a baseline liner. Nevertheless, the sensitivity magnitudes (approximately 7–9% under the refinement and substitution scopes explored) remain well bounded relative to observed between-design differences for CIP-Alpha concept variants that often exceed 15–20% in MPS95 reductions under oblique impacts.

The impact of representation choices underscores a practical lesson: when a slip interface is present and contributes materially to rotational kinematics attenuation, careful measurement and implementation of frictional behavior and interface compliance are necessary and beneficial. The observed 6% change in rotational acceleration under an HHS friction-law swap is significant but smaller than the gains imparted by the dedicated low-friction interface, supporting the interpretation that, while details of the friction law matter, the presence of the slip layer as a design element dominates the performance signature. For BDM, the contrast between viscohyperelastic and linear viscoelastic brains highlights the importance of nonlinear stiffness at large deformations and of accurate relaxation spectra at impact-relevant time scales.

In terms of validation relevance, the chosen matrix proved effective at bracketing the intended exposure and generated a rich set of observables that map directly to outputs. This directly benefited both HHS and, via boundary conditions, BDM. Additional value came from repeated trials at mid-severity levels, which helped stabilize central estimates and reduce confounding due to random variability, thereby clarifying whether observed differences between design variants and models were material or within noise.

Process controls and traceability were critical enablers. The inability to reproduce a figure or table rapidly undermines confidence; conversely, the documented zero unexpected diffs across the regression suite over an extended period, combined with immutable container images and repository tags, allowed independent internal review teams to replicate key results from scratch. This level of process discipline elevated the overall credibility posture beyond what would be achievable with ad hoc analysis pipelines.

Across all studied cases, halving characteristic element edge lengths (HHS 2.51.8 mm; BDM 2.01.4 mm, brainstem 1.51.0 mm) and reducing the explicit time step safety factor altered HHS peak resultant linear acceleration by 3.1% for oblique and 2.4% for near-normal drops and peak resultant angular acceleration by 5.2% and 4.0%, respectively; for BDM, MPS95 shifted by 6.7% (oblique) and 7.9% (near-normal), and 95th percentile strain rate changed by 7.3% (oblique) and 8.4% (near-normal), indicating that the adopted grids resolve outputs

of interest within decision-relevant tolerances.

19. Limitations and future work

The present credibility assessment has several limitations. First, while we covered two major impact mechanisms with speeds and angles reflective of the intended exposure domain, other mechanisms exist—such as highly eccentric glancing blows or complex multi-contact scenarios—that were not explicitly tested and could introduce additional sensitivity. Second, brain-model material parameters were drawn from literature and tuned to match representative dynamic moduli; in vivo conditions, anisotropy, microstructural heterogeneity, and potential rate effects beyond those captured remain sources of uncertainty. Incorporating uncertainty quantification (UQ) around these parameters and propagating to outputs would elevate the credibility level for the BDM and provide formal margins on strain and strain-rate estimates.

Third, the discretization studies, while targeted and reasonably comprehensive for the selected outputs, did not include a formal observed-order-of-accuracy analysis or a generalized GCI framework across all conditions; this choice reflected the practical challenges of asymptotic behavior in strongly nonlinear, contact-dominated problems and computational cost constraints at high resolutions. Nonetheless, targeted further refinement around peak-deformation regions and adaptive meshing strategies could reduce sensitivity. Fourth, the current HHS treats the skull as rigid, which is likely appropriate for the time scales and frequencies of interest but precludes exploring the potential coupling of skull flexure with scalp shear in rare conditions.

Fifth, while the alignment of laboratory conditions to simulations was strong for primary drivers and contact properties, we did not explore headform variability, helmet fit variability beyond a standardized protocol, or inter-laboratory differences. A multi-lab study with different headforms and shells would strengthen external validity and expose model robustness to perturbations in boundary conditions that arise in practice. Finally, while software process controls were comprehensive, formal external audits were not performed within the present cycle; future cycles will incorporate independent reviews and artifact signing to further enhance trust.

20. Conclusions

We assessed the credibility of two explicit finite element models used in the development and evaluation of the CIP-Alpha cranial impact protector. The HHS model, focused on skull and interface kinematics, demonstrated high solver fidelity, modest mesh sensitivity, tight mapping from laboratory inputs, bounded sensitivity to plausible representation alternatives, strong alignment of validation activities with intended use, and rigorous process control. The BDM, focused on brain deformation, exhibited similar solver fidelity and input alignment but greater sensitivity to mesh refinement and constitutive choices, supporting conservative grading for those elements.

In aggregate, the evidence supports a low–medium model risk posture for the stated context of use: pre-market design

verification of skull and brain kinematics under oblique and near-normal impacts and support for labeling claims confined to the tested severity and orientation envelope. The HHS is well suited to differentiate liner configurations and support statements about reductions in rotational and linear head kinematics, while the BDM is suitable to compare patterns and relative magnitudes of brain strain with the caveat that absolute values carry method- and mesh-dependent uncertainty on the order of several percent. Future work will target uncertainty quantification for tissue parameters, adaptive meshing, and expanded validation matrices to further elevate the BDM's credibility tier.

All inputs, meshes, and scripts were versioned under a controlled repository with immutable container images; 127 automated regression cases ran nightly for 180 consecutive days with zero unexpected diffs, and every model artifact in this paper resolves to a unique hash, ensuring reproducibility and auditability.

21. Acknowledgments

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22. References

References include published literature on viscohyperelastic modeling of brain tissue, explicit dynamics solver verification, helmet impact biomechanics, and friction characterization of liner–skin analogs, available upon request or in the supplementary materials archive associated with this assessment.

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Table 1: Credibility factors for HHS across oblique and normal impact mechanisms.

Credibility factor	Level	Basis
Numerical code verification - HHS - oblique rigid-anvil impact	4	Elastic wave phase error 0.3%; Taylor bar length error 0.7%; total energy drift 0.12% over 50 ms; momentum conserved within 0.2%.
Numerical code verification - HHS - normal compliant-anvil drop	4	Same solver stack: 0.3% wave error; 0.7% Taylor bar error; 0.12% energy drift; <0.2% momentum deviation in near-normal drops.
Discretization - HHS - oblique rigid-anvil impact	3	Refining helmet–liner–scalp from ~2.5 to 1.8 mm changed PLA by 3.1% and peak resultant angular acceleration by 5.2%; $\Delta\omega$ shift 2.8%.
Discretization - HHS - normal compliant-anvil drop	3	Same refinement changed PLA by 2.4% and peak resultant angular acceleration by 4.0%; $\Delta\omega$ shift 2.1%.
Equivalency of input parameters - HHS - oblique rigid-anvil impact	3	Inbound speed mismatch 1.4%; incident angle within 0.8°; CG alignment within 2 mm; scalp–liner $\mu = 0.05 \pm 0.01$.
Equivalency of input parameters - HHS - normal compliant-anvil drop	3	Inbound speed mismatch 1.2%; angle within 0.8°; CG within 2 mm; shell–anvil $\mu = 0.20 \pm 0.04$ (compliant).
Model form - HHS - oblique rigid-anvil impact	3	Switching to hyperfoam and Coulomb slip changed peak rotational acceleration by +6.0%; interface slip distance increased as expected.
Model form - HHS - normal compliant-anvil drop	3	Foam law substitution increased PLA by +4.2%; $\Delta\omega$ change <3%; Coulomb slip had negligible effect in near-normal contact.
Relevance of validation activities to the COU - HHS - oblique rigid-anvil impact	3	Speeds 3.3–6.5 m/s at 30°, 45°, 60° on rigid surfaces cover 78–92% of field severities/angles; observables map directly to outputs.
Relevance of validation activities to the COU - HHS - normal compliant-anvil drop	3	Near-normal drops onto compliant anvils at 3.3–6.5 m/s reflect product exposure; repeated trials at mid-severity stabilize estimates.
Software quality assurance - HHS - oblique rigid-anvil impact	4	127 nightly regression cases for 180 days with 0 unexpected diffs; container image tag v2026.3; artifacts pinned by hash.
Software quality assurance - HHS - normal compliant-anvil drop	4	Same SQA coverage with immutable artifacts; CI gate enforces tolerances; full traceability to solver image and input hashes.

Table 2: Credibility factors for BDM across oblique and normal impact mechanisms.

Credibility factor	Level	Basis
Numerical code verification - BDM - oblique rigid-anvil impact	4	Elastic wave phase error 0.3%; Taylor bar length error 0.7%; total energy drift 0.12% over 50 ms; momentum within 0.2%.
Numerical code verification - BDM - normal compliant-anvil drop	4	Benchmarks as above; production runs respect <0.5% energy-drift budget and <5% hourglass energy fraction.
Discretization - BDM - oblique rigid-anvil impact	2	Refining brain mesh from ~2.0 to 1.4 mm (brainstem 1.51.0 mm) changed MPS95 by 6.7% and strain-rate P95 by 7.3%.
Discretization - BDM - normal compliant-anvil drop	2	Same refinement changed MPS95 by 7.9% and strain-rate P95 by 8.4% under near-normal compliant impacts.
Equivalency of input parameters - BDM - oblique rigid-anvil impact	3	Boundary skull kinematics from HHS matched tests with 1.4% speed and 0.8° angle mismatch; CG within 2 mm.
Equivalency of input parameters - BDM - normal compliant-anvil drop	3	Boundary inputs matched within 1.2% speed and 0.8° angle; shell–anvil friction 0.20 ± 0.04 informs HHS and thus BDM BCs.
Model form - BDM - oblique rigid-anvil impact	2	Swapping to linear viscoelastic brain increased MPS95 by +9.1% and strain-rate P95 by +8.6% at 4.9 m/s, 45° oblique.
Model form - BDM - normal compliant-anvil drop	2	Linear viscoelastic substitution raised MPS95 by +8.7% and strain-rate P95 by +8.0% at 5.2 m/s near-normal on compliant anvil.
Relevance of validation activities to the COU - BDM - oblique rigid-anvil impact	3	Oblique validation spans 3.3–6.5 m/s and 30°–60°; skull kinematics directly validated, which drive BDM boundary conditions.
Relevance of validation activities to the COU - BDM - normal compliant-anvil drop	3	Near-normal compliant impacts included with repeated mid-severity trials; boundary kinematics map cleanly into BDM.
Software quality assurance - BDM - oblique rigid-anvil impact	4	127-case nightly regressions, 180 days, 0 unexpected diffs; all BDM inputs and outputs tie to immutable tags and hashes.
Software quality assurance - BDM - normal compliant-anvil drop	4	Identical SQA and CI enforcement as HHS; reproducibility verified from repository tag and container hash.